

Understanding Problem and Data

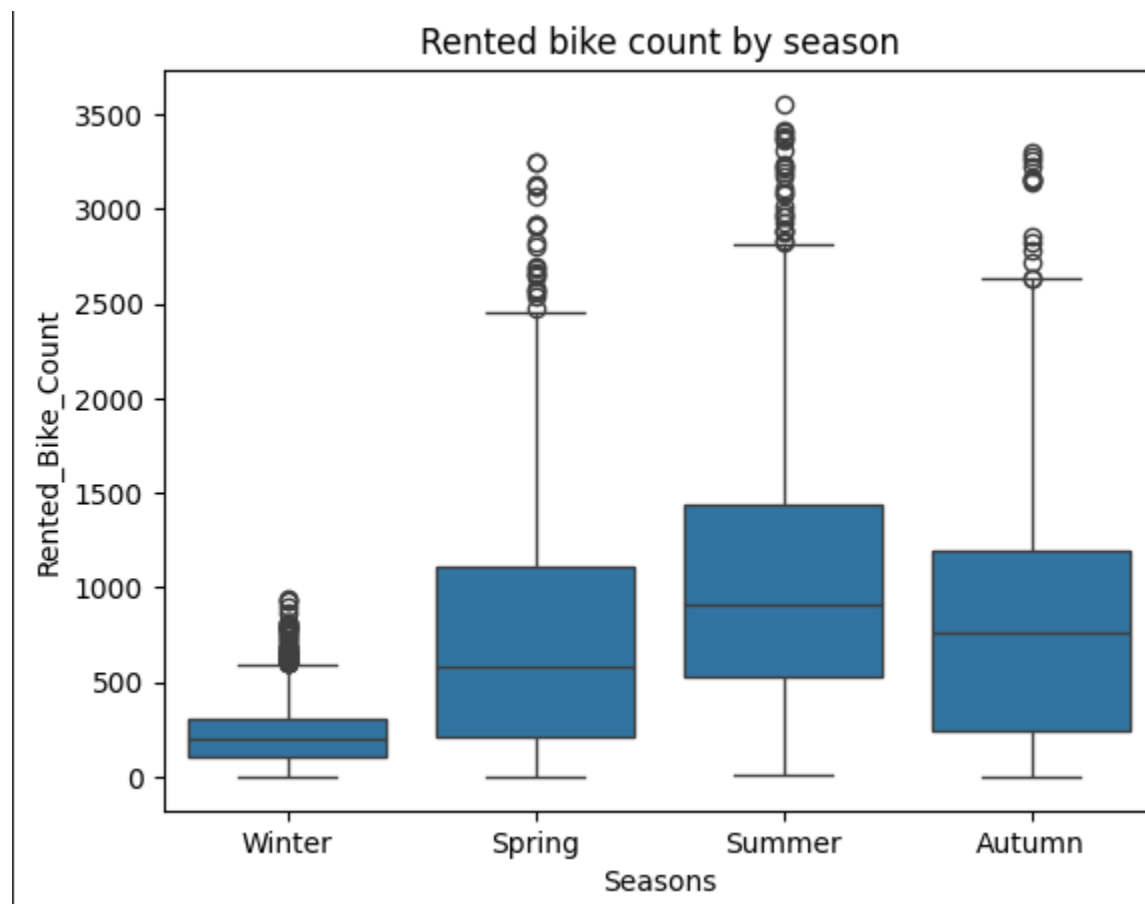
The dataset is from the UC Irvine Machine Learning Repository on bikes in Seoul.

Our Quantitative Predictor Variables are:

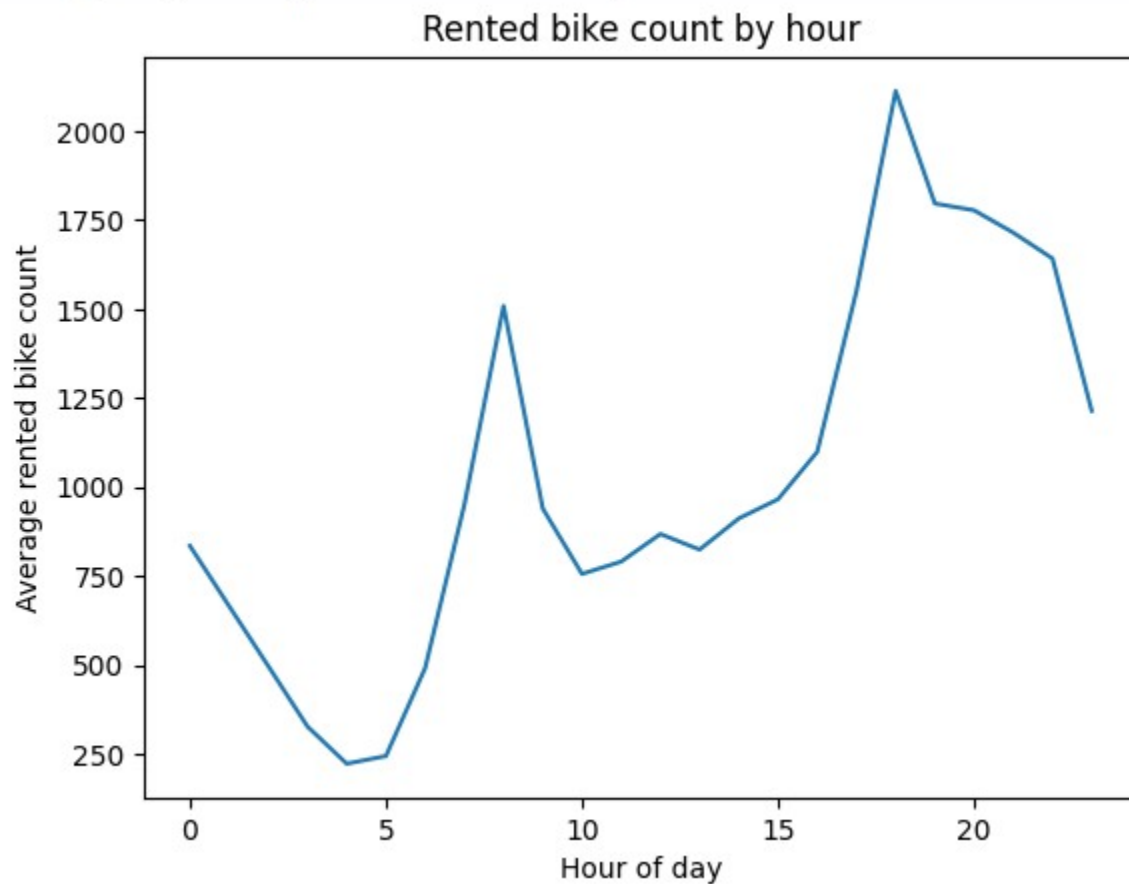
- Hour: Hour of the day
- Temperature: Temperature (Celsius)
- Humidity: Humidity Level (%)
- Wind_speed: windspeed (meters/second)
- Visibility: visibility in 10m
- Dew_point_temperature: dew point in temperature (Celsius)
- Solar_radiation: solar radiation (in MJ/m²; megajoules per square meter)
- Rainfall: amount of rainfall (mm)

All other variables will not be used for our analysis. We will also focus our analysis on the Summer season.

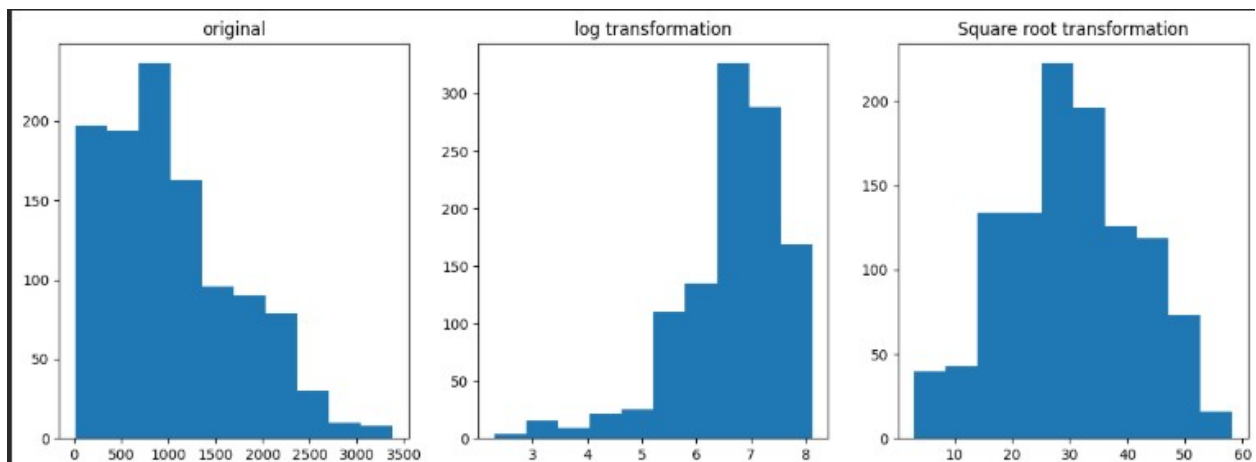
The goal of statistical analysis is to understand the utilization of public bicycles in the Seoul Bike Sharing System by understanding the number of bikes rented per hour (denoted by the variable Rented_Bike_Count)



We will be restricting our data to the summer season since this is the season that sees the most bike rentals compared to other seasons.



There are two big spikes in bike rentals during the summer at 8 AM and 6 PM.



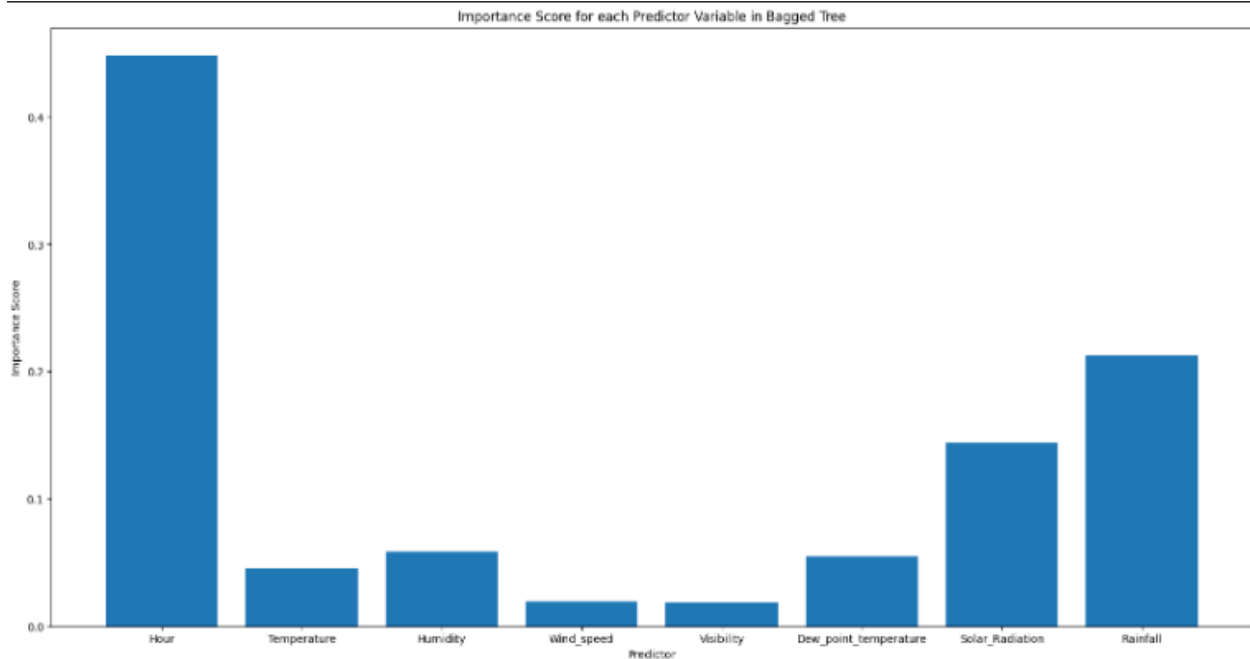
As shown above, we have transformed our response variable (Rented_Bike_Count) in order to see which one has the least skewed distribution. Our original data is skewed to the left, a log transformation is skewed to the right, however, a square root transformation achieves a much more normal distribution compared to the other two options. Therefore, we will proceed with a square root transformation for the remainder of our analysis.

Model Specification: Bagged Tree

The number of predictor variables we need to consider at each split of the decision tree is 8 (all of our variables). The parameters needed to be chosen for the modelling effort is the number of estimators.

Model Estimation: Bagged Tree

We use 5-fold cross-validation in order to find the $n_{\text{estimators}}$ for our model; we get 100.



As shown from the graph above, the most important variable when figuring out how many bikes are rented in Seoul during the Summer season is the hour of the day. The least important variable is visibility.

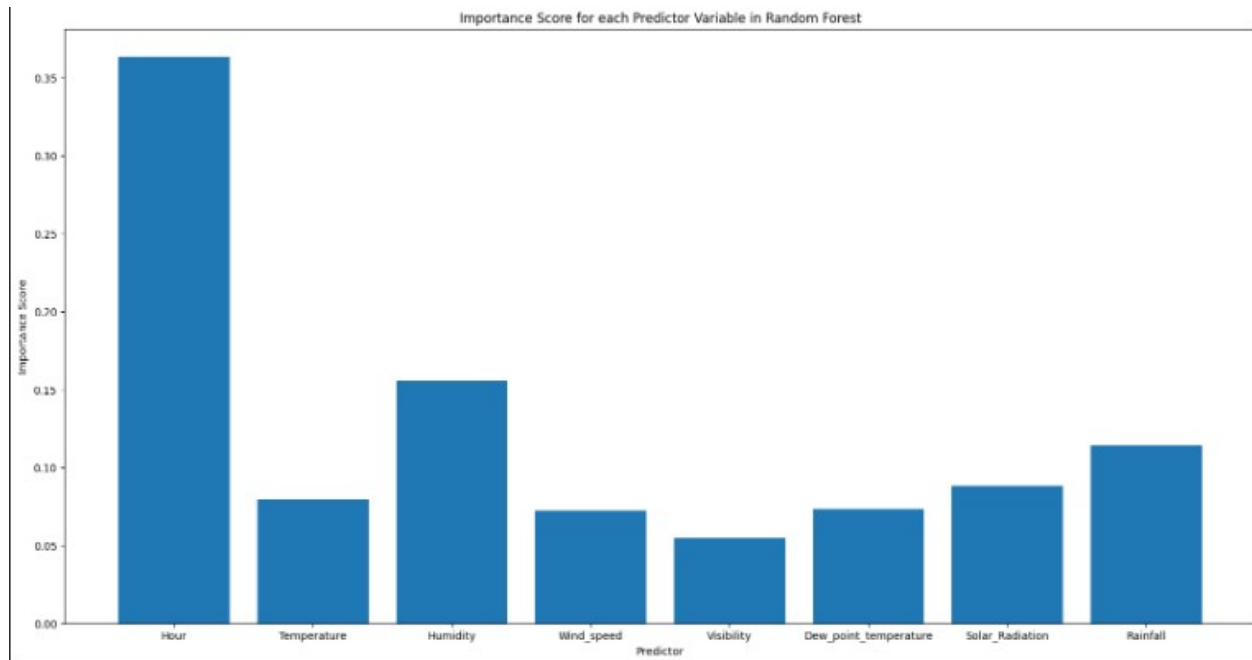
Model Specification: Random Forest

A random forest with $\text{max_features} = \text{"SQRT"}$ would create correlation amongst the trees, since the trees will most likely choose the same strong predictor variable early. Restricting each tree adds randomness to each tree. This also makes it much easier computationally, since if there are fewer predictor variables, machines need to do less work.

Parameters needed to be chosen for this modelling effort are the number of estimators.

Model Estimation: Random Forests

We use 5-fold cross-validation to find the $n_{\text{estimators}}$ for our model; we get 1000



As shown in the graph above, the most important and least important variables for figuring out the number of bikes rented in Seoul are the same as our previous Bagging tree model, Hours and Visibility. However, a big difference between the two graphs is that the variables are much more balanced, comparatively, than they were in the first graph in Model Estimation: Bagged Trees.

Model Specification: Boosted Tree

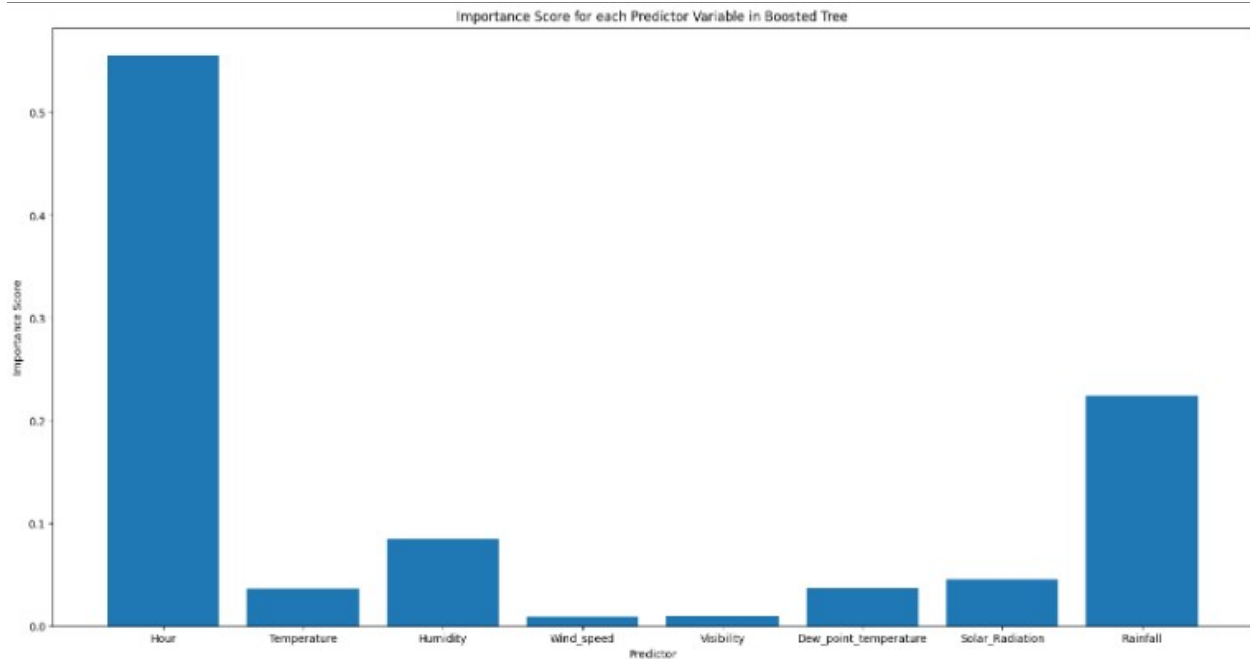
We consider a boosted tree where $\text{max_depth} = 1$. This means that each individual tree has only one split, uses one predictor variable at that split, and creates only two terminal nodes.

Parameters needed for this modelling effort would be $n_estimators$ and $learning_rate$

Model Estimation: Boosted Tree

We use 5-fold cross-validation to find our parameters; we get

- $N_estimators = 1000$
- $Learning_rate = 0.1$



The most important variable when determining the number of bikes rented in Seoul during the Summer season for our Boosted model is Hour, and the least important variable is Wind_Speed.

Model Assessment:

We will now report the mean squared error on the test data for each model

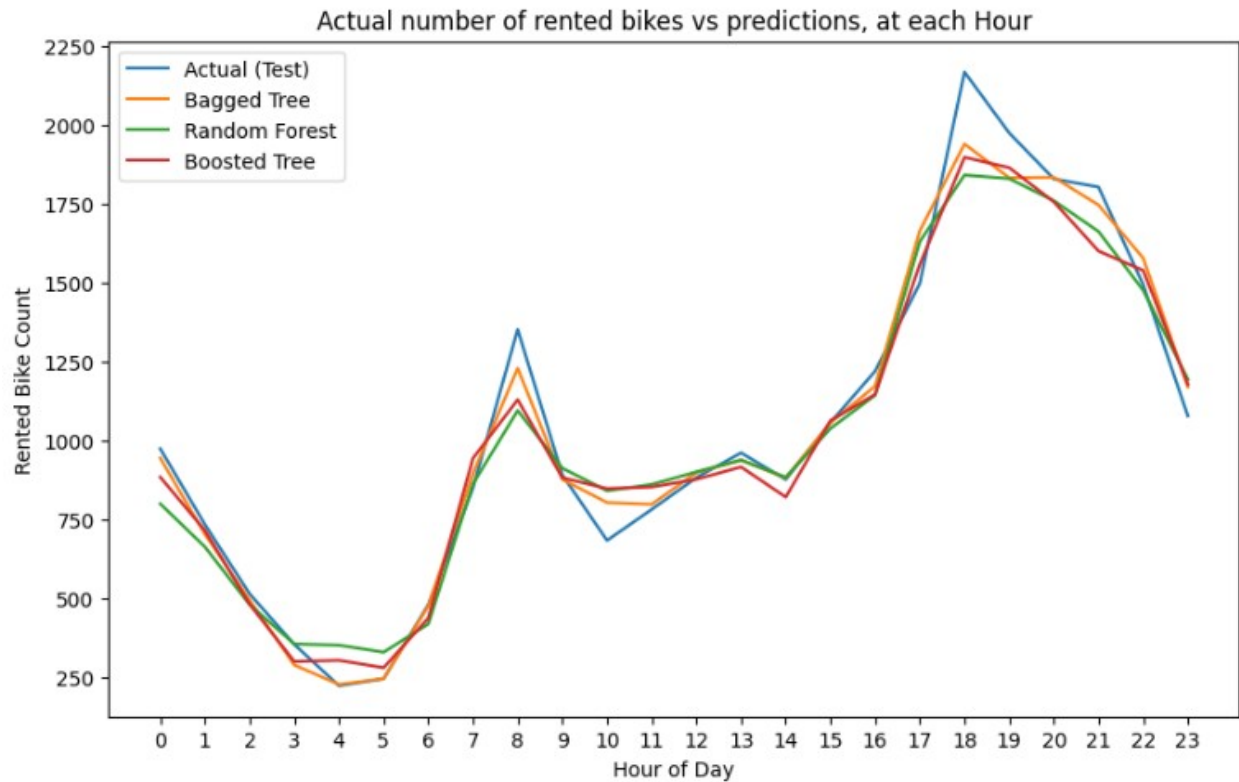
- Bagged Tree: 20.230
- Random Forest: 22.906
- Boosted Tree: 26.729

We recommend the Bagged Tree model since it achieved the lowest mean squared error compared to the Random Forest and the Boosted Tree model.

Model Deployment:

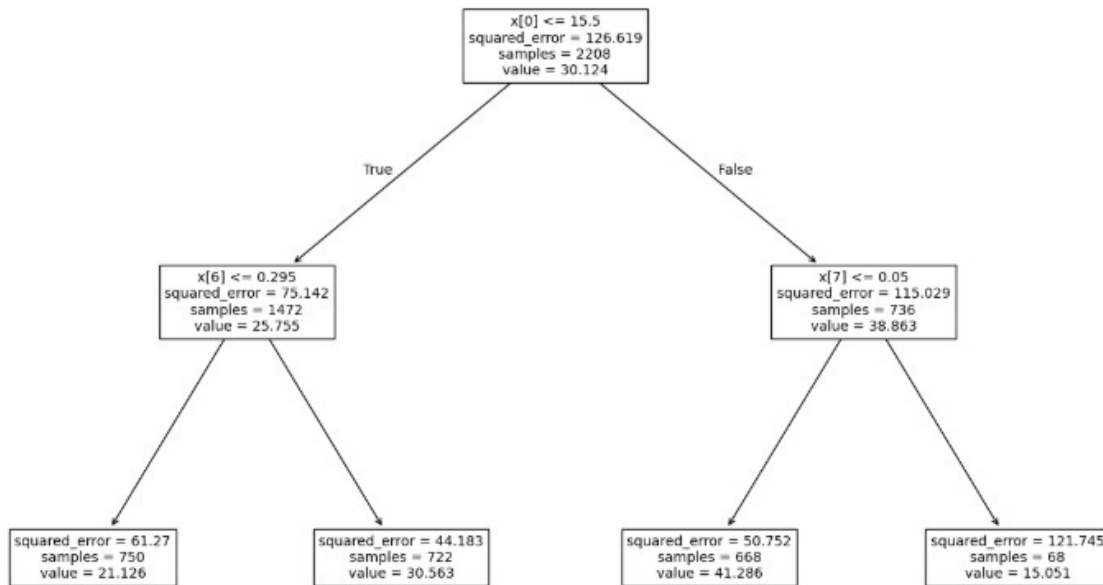
We consider a predictor profile that utilizes the mean of each predictor variable, except the hour of the day, for the full dataset of the summer season.

- Temperature: 12.883
- Humidity: 58.226
- Wind_speed: 1.723
- Visibility: 1436.826
- Dew_point_temperature: 4.074
- Solar_Radiation: 0.569
- Rainfall: 0.149



As shown in the graph above, we can see that each model has similar spikes in the number of rented bikes around the 8 AM and 6 PM mark. If we take a closer look, we can see that the bagged tree generally mimics the actual test data the best, compared to the other two models, especially around the two peaks, meaning that it performs the best on our test data relative to the other two models.

Decision Tree, 4 terminal nodes



For a more interpretable model, we built a single decision tree on the full dataset of the summer season with 4 terminal nodes. The tree makes decisions based on

Hour: $x[0]$
 Solar Radiation: $x[6]$
 and Rainfall: $x[7]$

Our first split comes from the hour of the day, with our condition being whether the hour of the day is less than 15.5 or greater than/ equal to 15.5.

Our other split comes from Solar Radiation, with the condition being whether the radiation is less than 0.295 MJ/m² or greater than/ equal to 0.295.

Our last split comes from RainFall, with the condition being whether the amount of rainfall in mm is less than 0.05 or greater than/ equal to 0.05.

Terminal Node	Expected Rentals	What can be said
1	ER = 21 Hours ≤ 15.5 Solar Radiation ≤ 0.295	Low sunlight and early morning can lead to relatively low numbers of rentals
2	ER = 31 Hours ≤ 15.5	More sunlight can increase bike demand during the day.

	Solar Radiation > 0.295	
3	ER = 41 Hours > 15.5 Rainfall <= 0.05	Low amounts of rain in the afternoon and evening can lead to an increase in bike rentals.
4	ER = 15 Hours > 15,5 Rainfall > 0.05	Rain has a substantial negative impact on bike rental demand.